

Magnetic Effects of Electric Current (Set B) — MCQ Worksheet

Science • Chapter 12 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Magnetic field is a:

- (A) Vector quantity (has direction and magnitude)
- (B) Scalar
- (C) Pure number
- (D) Force only

Q2. Magnetic field lines form:

- (A) Closed loops
- (B) Open lines
- (C) Crossing pairs
- (D) Spirals only

Q3. A compass needle placed near a current-carrying wire:

- (A) Deflects (showing magnetic field around current)
- (B) Burns
- (C) Melts
- (D) Has no effect

Q4. Magnetic field lines around a straight wire are:

- (A) Concentric circles
- (B) Straight lines
- (C) Parabolas
- (D) Random

Q5. Right-hand thumb rule gives:

- (A) Direction of magnetic field around a wire carrying current
- (B) Force on conductor
- (C) Induced EMF
- (D) Voltage

Q6. A solenoid behaves like a:

- (A) Bar magnet
- (B) Resistor
- (C) Capacitor
- (D) Battery

Q7. Adding soft iron core to solenoid:

- (A) Increases field strength (makes electromagnet)
- (B) Decreases field
- (C) Has no effect
- (D) Reverses field

Q8. Force on current-carrying conductor in magnetic field is given by Fleming's:

- (A) Left-hand rule
- (B) Right-hand rule
- (C) Up-thumb rule
- (D) Down-thumb rule

Q9. In Fleming's left-hand rule, the middle finger represents:

- (A) Current
- (B) Force
- (C) Magnetic field
- (D) Voltage

Q10. An electric motor converts:

- (A) Electrical energy into mechanical
- (B) Mechanical into electrical
- (C) Chemical into electrical
- (D) Light into electrical

Q11. Split-ring commutator in DC motor:

- (A) Reverses current direction in coil every half rotation
- (B) Stops current
- (C) Increases voltage
- (D) Filters AC

Q12. Electric generator works on principle of:

- (A) Electromagnetic induction
- (B) Heating effect
- (C) Magnetic effect of current
- (D) Chemical effect

Q13. Direction of induced current in a generator follows:

- (A) Fleming's right-hand rule
- (B) Fleming's left-hand rule
- (C) Ohm's law
- (D) Faraday's first law

Q14. AC frequency in India is:

- (A) 50 Hz
- (B) 60 Hz

(C) 100 Hz

(D) 25 Hz

Q15. Live wire is:

(A) Red

(B) Black

(C) Green

(D) Yellow

Q16. Two parallel wires carrying current in same direction:

(A) Attract each other

(B) Repel each other

(C) Have no force

(D) Become electrolytes

Q17. Two parallel wires carrying current in opposite direction:

(A) Repel each other

(B) Attract each other

(C) Have no force

(D) Get magnetised

Q18. Earthing in household wiring:

(A) Provides a low-resistance path to earth for leakage currents

(B) Increases voltage

(C) Reduces shock chance to zero always

(D) Generates current

Q19. If current flows from north to south in a horizontal wire above a compass, the north of compass:

(A) Deflects to east (using right-hand rule)

(B) Deflects to west

(C) Points up

(D) No deflection

Q20. Why is the soft iron core preferred in electromagnets?

(A) High permeability, easily magnetised and demagnetised

(B) Cheap

(C) Heavy

(D) Cold

Q21. If number of turns in a solenoid is doubled keeping current same, magnetic field strength:

(A) Approximately doubles

(B) Halves

(C) Stays same

(D) Becomes zero

Q22. Galvanometer is used to:

(A) Detect and measure small electric currents

(B) Voltage only

(C) Resistance only

(D) Power

Q23. An AC generator differs from DC by using:

(A) Slip rings instead of split rings

(B) More turns

(C) Larger magnet

(D) Stronger battery

Q24. Why is electric shock from AC more dangerous than DC at same voltage?

(A) AC causes muscles to grip; can't release wire

(B) DC is invisible

(C) AC is hotter

(D) AC is faster

Q25. A current-carrying coil placed in a magnetic field experiences:

(A) A torque (turning force) about its axis

(B) No force

(C) Heat only

(D) Magnetic field of own